



Problem

0 → Empty cell 1 → Fresh orange 2 → Rotten orange

A rotten orange can rot an adjacent fresh orange in 1 unit of time.

A rotten orange at position (i, j) can rot fresh oranges at:

(i-1, j) → Up (i+1, j) → Down (i, j-1) → Left (i, j+1) → Right

Find the minimum time required to rot all fresh oranges.

If it is impossible to rot all fresh oranges, print -1.

If there are no fresh oranges initially, the required time is 0.

Input Format

The first line contains an integer T, the number of test cases. For each test case: The first line contains two integers N and M, representing the number of rows and columns. The next N lines contain M space-separated integers representing the grid.

Constraints

$1 \leq T \leq 100$ $1 \leq N, M \leq 500$ $\text{grid}[i][j] \in \{0, 1, 2\}$

Output Format

Print a single integer representing the minimum time required to rot all fresh oranges.

Print -1 if some fresh oranges can never become rotten

Sample Input 0

```
3 3
2 1 1
1 1 0
0 1 1
```

Sample Output 0

```
4
```

Sample Input 1

```
3 3
2 1 1
0 1 1
1 0 1
```

Sample Output 1

Change Theme

Language

Python 3



```
1 # Enter your code here. Read input from STDIN. P
2 from collections import deque
3 n,m=map(int,input().split())
4 a=[]
5 for i in range(n):
6     a.append(list(map(int,input().split())))
7 q=deque()
8 fresh=0
9 for i in range(n):
10     for j in range(m):
11         if a[i][j]==2:
12             q.append((i,j))
13         elif a[i][j]==1:
14             fresh+=1
15 time=0
16 while q:
17     size=len(q)
18     for i in range(size):
19         x,y=q.popleft()
20         if x>0 and a[x-1][y]==1:
```

Line: 41 Col: 14



Upload Code as File

Run Code

Submit Code



Test against custom input



Test case 0



Test case 1

Compiler Message

Success